

Program : Diploma in Chemical Engineering	
Course Code : 5073A	Course Title: Petroleum Refining & Petrochemical Technology
Semester : 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L: 4 T: 0 P: 0)	Periods per semester: 60

Course Objectives:

- To impart the basic knowledge on fossil fuels, up-stream and downstream processes for crude petroleum.
- To comprehend the manufacturing processes and applications of various petrochemical products.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Unit Operations		Chemical Process Principles	2

Course Outcomes :

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Summarise fossil fuels and the up-stream processes of crude petroleum.	14	Understanding
CO2	Illustrate the downstream processes for crude petroleum.	17	Understanding
CO3	Summarize the treatment techniques for petroleum fractions.	15	Understanding
CO4	Summarize the manufacturing processes and applications of various petrochemicals.	14	Understanding

CO – PO Mapping

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	2						
CO2	2						
CO3	2						
CO4						3	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Summarise fossil fuels and the up-stream processes of crude petroleum.		
M1.01	Classify fossil fuels.	2	Remembering
M1.02	Summarize the production methods, characteristics and properties of gaseous fuels.	4	Understanding
M1.03	Summarise the characterization and properties of crude petroleum.	4	Understanding
M1.04	Illustrate the upstream processes for crude petroleum.	4	Understanding
Contents: Classification of fuel- Categorize primary & secondary solid, liquid and gaseous fuels with examples. Identify the composition, properties and uses of Natural gas, - Illustrate the production of natural gas - Comprehend the production of shale gas. Origin& formation of petroleum - Classification of crude oil – Properties of crude oil -API gravity and density, viscosity, sulphur content and vapour pressure – Evaluation of crude oil. Exploration and drilling of crude oil - Gas-oil separation – Dehydration - Desalting- Crude stabilization and Enhanced oil recovery – Sources of indigenous crude - Sources of imported crude - Petroleum industries in India			
CO2	Illustrate the downstream processes for crude petroleum.		
M2.01	Illustrate modern refinery operations.	1	Understanding
M2.02	Illustrate petroleum products from crude oil by fractional distillation.	4	Understanding
M2.03	Outline various cracking and reforming processes of petroleum fractions	4	Understanding

M2.04	Summarise Visbreaking, Coking, hydro treating processes.	4	Understanding
M2.05	Summarise Alkylation, Isomerisation and polymerization processes	4	Understanding
	Series Test – I		
Contents: Block diagram of a modern refinery operation (Including ADU, VDU, gas processing, hydrotreater, FCCU, isomeriser, alkylation, blending, DCU and sulphur recovery unit) Illustrate Crude oil fractionation by distillation- crude preheater- atmospheric distillation unit- vacuum distillation unit-petroleum products - List the petroleum products from crude oil refining and their boiling ranges. Illustrate: Cracking - hydro cracking, thermal cracking, catalytic cracking, fixed bed and fluidized bed - Catalytic reforming Illustrate: Visbreaking operation – Coking - hydro treating Illustrate: Sulphuric acid Alkylation process, Low temperature Isomerisation process and Catalytic polymerization processes.			
CO3	Summarize the treatment techniques for petroleum fractions.		
M3.01	Identify the physical and chemical impurities in petroleum fractions.	2	Understanding
M3.02	Summarize the purification and treatment processes of petroleum fractions.	6	Understanding
M3.03	Summarize the important properties of petroleum products.	2	Understanding
M3.04	Outline the experimental methods for determining the properties of petroleum products.	5	Understanding
Contents: Physical and chemical impurities in petroleum fractions. Purification and treatment process - sweetening process of petroleum fractions - LPG Merox treatment - hydrodesulphurization of gasoline, kerosene, and diesel – treatment for lube oil. Properties of petroleum fractions (API gravity - Watson Characterization factor - Viscosity - Sulfur content - True boiling point (TBP) curve - Pour point - Flash and fire point - ASTM distillation curve - Knocking -Octane number, Cetane number - Smoke point) Experimental methods for the measurement of viscosity, flash and fire point, cloud point, pour point.			
CO4	Summarize the manufacturing processes and applications of various petrochemicals.		
M4.01	Outline the feedstock for petrochemicals.	4	Understanding

M4.02	Summarize the applications of petrochemicals	3	Understanding
M4.03	Illustrate the chemicals produced from methane, synthesis gas and olefins.	3	Understanding
M4.04	Comprehend the chemicals from aromatics.	4	Understanding
	Series Test – II		

Contents:

Feedstock for petrochemicals – List the major Gas, Liquid and Solid feedstock's for petrochemical industry - Identify the products from above feedstock's.

List the major applications of petrochemicals - polyethylene, polypropylene, polyester, polybutadiene, polyacrylate.

Production of methanol by oxidation of methane – Methanol from synthesis gas – production of formaldehyde from methanol. Illustrate the manufacturing of ethylene and propylene by thermal cracking – manufacturing of polyethylene - List the propylene derivatives – Illustrate the production of Acrylic Acid and Methyl Methacrylate (MMA).

Illustrate the Manufacturing process for Phenol, Polystyrene

Group activity (Prepare a report on the manufacturing of Caprolactam in FACT and compare the production method with other available manufacturing methods)

Text / Reference

T/R	Book Title/Author
T1	Bulk Chemicals from Petroleum; B.K. Bhaskarao
R1	Modern Petroleum Refining Processes; B. K. Bhaskarao
R2	Petroleum Refinery Engineering; W.L. Nelson,
R3	Elements of Petroleum Refinery Engineering; O.P. Gupta
R4	Elements of Petrochemical Engineering; Saikat Maitra & O.P. Gupta

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/103/102/103102022/